

HISSAR, India

WMO: 421310

Lat: 29.1736N

Lon: 75.7393E

Elev: 213

StdP: 98.79

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.0	6.3	0.3	3.9	18.8	2.1	4.5	17.0	3.1	17.4	2.5	16.6	0.4	320	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.5	44.5	23.7	43.1	23.8	41.6	23.9	29.0	35.4	28.6	34.8	28.2	34.2	1.4	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.7	24.3	31.8	27.2	23.6	31.4	26.9	23.2	31.2	97.1	35.8	95.0	34.8	93.1	34.3	35.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
3.2	2.5	2.3	DB	3.4	46.7	2.0	0.9	2.0	47.3	0.8	47.9	-0.3	48.4	-1.7	49.0	
			WB	2.6	30.0	1.8	1.2	1.3	30.9	0.2	31.5	-0.8	32.2	-2.2	33.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.8	13.5	17.5	23.3	30.0	33.8	34.1	32.5	31.3	30.3	27.0	21.1	15.1
	DBStd	7.67	2.58	3.03	3.46	3.41	3.06	3.23	2.50	2.04	1.96	2.54	2.44	2.87
	HDD10.0	5	3	0	0	0	0	0	0	0	0	0	0	2
	HDD18.3	311	152	47	3	0	0	0	0	0	0	0	4	104
	CDD10.0	5784	112	211	413	601	739	724	696	660	610	528	332	160
	CDD18.3	3049	2	25	158	351	481	474	438	402	360	269	86	5
	CDH23.3	42439	9	202	1600	4937	7695	7589	6522	5624	4686	2855	680	39
Wind	WSAvg	0.7	0.6	0.7	0.8	0.9	0.9	1.0	0.9	0.8	0.7	0.4	0.3	0.4
	PrecAvg	463	12	17	16	13	23	56	125	103	84	12	2	4
	PrecMax	772	50	90	101	99	83	175	476	259	232	142	13	28
	PrecMin	99	2	0	0	0	0	0	0	0	4	0	0	0
Precipitation	PrecStd	175	12	22	23	21	24	47	109	74	68	32	4	7
	DB	24.7	30.2	37.8	43.9	46.2	46.0	42.6	39.9	38.8	37.6	32.1	26.2	
	MCWB	17.0	18.9	21.1	21.7	23.9	23.9	26.9	27.4	25.9	23.5	19.6	16.7	
	DB	22.2	27.6	35.0	42.1	44.8	44.4	40.8	38.3	37.6	36.1	30.5	24.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	14.8	17.8	20.7	21.3	23.3	24.6	27.1	27.4	25.7	23.4	19.1	15.9	
	DB	20.7	25.6	32.8	40.4	43.5	42.9	39.2	36.9	36.6	34.8	29.1	23.0	
	MCWB	13.9	17.0	20.0	21.2	23.4	24.8	27.3	27.5	25.6	22.9	18.7	15.0	
	DB	19.2	23.8	30.9	38.6	41.9	41.3	37.6	35.6	35.5	33.5	27.4	21.5	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCWB	13.0	16.2	19.4	20.8	23.2	25.3	27.5	27.6	25.7	22.3	18.2	14.3	
	WB	17.8	20.5	23.5	25.1	27.9	29.1	29.6	29.5	28.7	26.9	21.3	18.4	
	MCDB	22.7	27.1	32.3	38.2	40.3	38.1	35.4	35.0	34.4	34.0	28.9	23.6	
	WB	16.1	19.1	22.1	23.5	26.8	28.3	29.1	28.9	28.1	25.7	20.4	17.1	
Mean Daily Temperature Range	MCDB	20.3	25.5	31.3	37.2	39.2	36.9	35.4	34.5	33.7	32.2	27.9	22.5	
	WB	15.0	18.1	21.1	22.5	25.9	27.8	28.7	28.5	27.6	24.7	19.6	16.0	
	MCDB	18.9	23.8	30.1	36.1	38.0	36.0	34.9	33.9	32.9	31.1	27.0	21.3	
	WB	14.0	17.0	20.3	21.6	24.9	27.3	28.3	28.1	27.1	23.9	18.9	15.1	
Clear-Sky Solar Irradiance	MCDB	17.6	22.2	28.8	35.1	37.0	35.7	34.2	33.4	32.2	30.5	26.0	20.0	
	MDBR	10.9	12.4	14.0	15.6	14.7	11.5	8.3	7.8	9.7	13.2	13.5	11.9	
	5% DB	13.2	14.1	16.1	16.9	16.1	13.9	10.3	9.6	11.4	14.0	14.6	14.0	
	MCDBR	6.3	5.7	5.1	4.8	4.7	3.8	2.7	3.0	3.3	4.4	5.6	6.4	
All-Sky Solar Radiation	MCWBR	11.0	12.6	15.0	15.2	13.8	10.8	8.7	8.3	9.0	11.6	12.9	12.2	
	MCWBR	6.1	5.8	5.7	5.5	5.1	3.7	3.0	3.1	3.4	4.5	5.5	6.1	
	taub	0.485	0.463	0.492	0.568	0.668	0.748	0.766	0.655	0.572	0.644	0.606	0.533	
	taud	1.852	1.968	1.924	1.778	1.619	1.521	1.522	1.733	1.867	1.614	1.643	1.756	
Historical Trends	Ebn at Noon	717	791	798	752	680	625	611	678	721	620	597	642	
	Edh at Noon	174	168	187	223	263	288	287	231	196	237	214	184	
	RadAvg	3.11	4.33	5.64	6.57	6.84	6.08	5.31	5.32	5.33	4.69	3.63	3.11	
	RadStd	0.34	0.32	0.24	0.24	0.22	0.40	0.38	0.34	0.31	0.22	0.24	0.20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page